

Smart Storage Behavioural Intelligence System Using IoT and Machine Learning for Anomaly Detection and Real-Time Security Alerts

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Abstract

Conventional storage security systems primarily rely on mechanical locking mechanisms or basic intrusion detection methods, which are often insufficient to identify abnormal access behaviour and provide timely remote notifications. This paper proposes a Smart Storage Behavioural Intelligence System Using IoT and Machine Learning for Anomaly Detection and Real-Time Security Alerts. The proposed system employs an ESP32 microcontroller integrated with a magnetic reed switch, relay-controlled solenoid lock, DS3231 real-time clock, Firebase Realtime Database, and a WebSocket-based communication framework to monitor and record storage access activities. Access parameters such as access time, door open duration, and access frequency are continuously collected and stored in the cloud for behavioural analysis. An Isolation Forest machine learning algorithm is trained using historical access records to learn normal user behaviour and identify suspicious access patterns as anomalies. Whenever an abnormal event is detected, the system immediately generates real-time Telegram notifications and activates a local buzzer to alert the user. The integration of embedded hardware, cloud computing, and behavioural intelligence provides a cost-effective, scalable, and intelligent security solution suitable for lockers, offices, residential storage units, and other smart storage applications. Experimental observations demonstrate that the proposed system effectively distinguishes normal and abnormal access behaviour while enabling reliable real-time monitoring and rapid security response.

Keywords: *Internet of Things (IoT), Smart Storage System, Behavioural Intelligence, Machine Learning, Isolation Forest, Anomaly Detection, ESP32, Firebase Realtime Database, WebSocket Communication, Telegram Alerts, Real-Time Security Monitoring*

1. Introduction

The rapid advancement of the Internet of Things (IoT) has significantly transformed conventional security systems by enabling intelligent monitoring, remote accessibility, and real-time decision-making. Smart storage systems have become increasingly important for protecting valuable assets in residential, commercial, industrial, and institutional environments. However, many existing storage security solutions

rely primarily on mechanical locks, password-based authentication, or simple motion sensors, which are unable to analyse user behaviour or detect suspicious access patterns. As a result, unauthorized access may remain undetected until after a security breach has occurred. Recent developments in machine learning have enabled security systems to move beyond traditional rule-based detection methods by learning normal behavioural patterns and identifying unusual activities automatically. Behaviour-based anomaly detection provides an effective approach for recognizing abnormal access without requiring predefined attack signatures. This capability is particularly beneficial for smart storage applications, where access behaviour varies depending on the user, time, and frequency of usage.

This paper proposes a Smart Storage Behavioural Intelligence System Using IoT and Machine Learning for Anomaly Detection and Real-Time Security Alerts. The proposed system integrates an ESP32 microcontroller, magnetic reed switch, relay-controlled solenoid lock, DS3231 Real-Time Clock (RTC), Firebase Realtime Database, WebSocket communication, and an Isolation Forest machine learning algorithm to continuously monitor storage access activities. The system records important behavioural parameters such as access time, door open duration, and access frequency, which are stored in a cloud database for behavioural analysis. By learning historical access patterns, the machine learning model can distinguish normal user behaviour from abnormal activities. Whenever an anomaly is detected, the system immediately generates real-time Telegram notifications and activates a local buzzer, enabling users to respond quickly to potential security threats. The proposed system combines embedded hardware, cloud-based data management, and intelligent behavioural analysis to provide a low-cost, scalable, and efficient smart storage security solution. Unlike conventional security systems that rely solely on fixed thresholds or manual monitoring, the proposed approach continuously adapts to user behaviour, thereby improving security, reducing false alarms, and supporting real-time remote monitoring for modern IoT-enabled storage applications.

2. Literature Survey

Recent advancements in the Internet of Things (IoT) and Machine Learning (ML) have significantly improved smart security systems by enabling intelligent monitoring and anomaly detection. Several researchers have proposed IoT-based storage security systems; however, limitations still exist in behavioural learning, cloud integration, and real-time alert mechanisms.

Existing IoT-based storage security systems primarily depend on authentication techniques such as RFID, passwords, or biometrics to prevent unauthorized access [4], [5]. Although these methods improve access control, they are unable to analyse user behaviour after authentication. Similarly, recent machine learning approaches have demonstrated promising performance in anomaly detection for IoT environments [1]–[3]; however, most studies focus on network intrusion detection or cybersecurity applications instead of behavioural monitoring in physical storage systems.

To address these limitations, the proposed Smart Storage Behavioural Intelligence System integrates ESP32 hardware, Firebase Realtime Database, WebSocket communication, and the Isolation Forest algorithm to continuously learn user access behaviour based on access time, door open duration, and access frequency. Unlike existing systems, the proposed solution performs real-time behavioural anomaly detection and immediately notifies users through Telegram alerts while simultaneously activating a local buzzer, thereby providing an intelligent, cloud-connected, and scalable smart storage security solution.

Ref.	Author(s)	Methodology	Advantages	Limitations
[1]	Alve et al.	IoT Security using Machine Learning	Improved intrusion detection accuracy	Focuses on network attacks rather than physical storage security
[2]	Sami and Abid	Isolation Forest for IoT Anomaly Detection	Lightweight and computationally efficient	Does not analyse user access behaviour in smart storage
[3]	Ripan et al.	Isolation Forest-based Cyber Anomaly Detection	Effective anomaly identification with low complexity	Limited to cybersecurity datasets and lacks hardware implementation
[4]	RFID-based Smart Locker System	RFID Authentication	Easy user authentication	Unable to learn behavioural patterns or detect suspicious usage
[5]	Password/Biometric Storage Systems	Authentication-based Access Control	Provides secure authentication	Does not provide behavioural analysis or intelligent anomaly detection

Table I. Comparative Analysis of Existing Research

3. Proposed System

The proposed Smart Storage Behavioural Intelligence System is designed to enhance the security of storage units by integrating Internet of Things (IoT) technology with machine learning-based behavioural analysis. Unlike conventional storage systems that rely only on physical locks or authentication mechanisms, the proposed system continuously monitors user access behaviour and detects suspicious activities in real time.

The system consists of an ESP32 microcontroller connected to a magnetic reed switch, a relay-controlled solenoid lock, a DS3231 Real-Time Clock (RTC), and a buzzer. The reed switch continuously monitors the door status, while the RTC records the exact time of every access event. The ESP32 collects behavioural parameters including access hour, door open duration, and access frequency. These parameters are transmitted to a Node.js server through WebSocket communication and are subsequently stored in the Firebase Realtime Database.

The collected historical access records are analysed using the Isolation Forest machine learning algorithm to learn normal user behaviour. Whenever a new access event is received, the trained model evaluates the behavioural pattern and classifies it as either Normal or Anomalous. If an anomaly is detected, the server immediately sends a Telegram notification to the authorized user and simultaneously activates a buzzer through the ESP32, thereby providing both remote and local security alerts.

The proposed architecture combines embedded hardware, cloud-based data storage, real-time communication, and intelligent behavioural learning into a unified security framework. This approach not only provides secure storage monitoring but also enables adaptive behavioural analysis, making the system suitable for residential lockers, office cabinets, industrial storage units, and other smart security applications

4. System Architecture.

The architecture of the proposed Smart Storage Behavioural Intelligence System consists of four major layers: the sensing layer, communication layer, cloud processing layer, and intelligent alert layer. These layers work together to provide continuous monitoring, behavioural analysis, and real-time security notifications.

At the sensing layer, an ESP32 microcontroller interfaces with a magnetic reed switch to detect the open or closed state of the storage door. A relay-controlled solenoid lock is used to provide authorized access, while a DS3231 Real-Time Clock (RTC) records the exact time of every access event. A buzzer is also connected to the ESP32 to generate local security alerts whenever suspicious behaviour is detected.

The communication layer establishes a Wi-Fi connection between the ESP32 and the Node.js server using the WebSocket protocol. The ESP32 continuously transmits behavioural parameters, including access hour, door open duration, and access frequency, to the server with low communication latency.

The cloud processing layer consists of a Node.js server connected to the Firebase Realtime Database. All access records received from the ESP32 are stored in Firebase to maintain historical behavioural information. The stored data is processed using the Isolation Forest machine learning algorithm implemented in Python. The model learns normal access behaviour from historical records and evaluates each new access event to determine whether it represents normal activity or an anomaly.

The intelligent alert layer provides immediate security notifications based on the prediction generated by the machine learning model. If the detected behaviour is classified as anomalous, the server simultaneously sends a Telegram notification to the authorized user and transmits an alert message to the ESP32 through the WebSocket connection. Upon receiving the alert, the ESP32 activates the buzzer to indicate a potential security breach. This integrated architecture enables real-time behavioural monitoring, cloud-based data management, intelligent anomaly detection, and rapid user notification, thereby improving the overall reliability and effectiveness of the proposed smart storage security system.

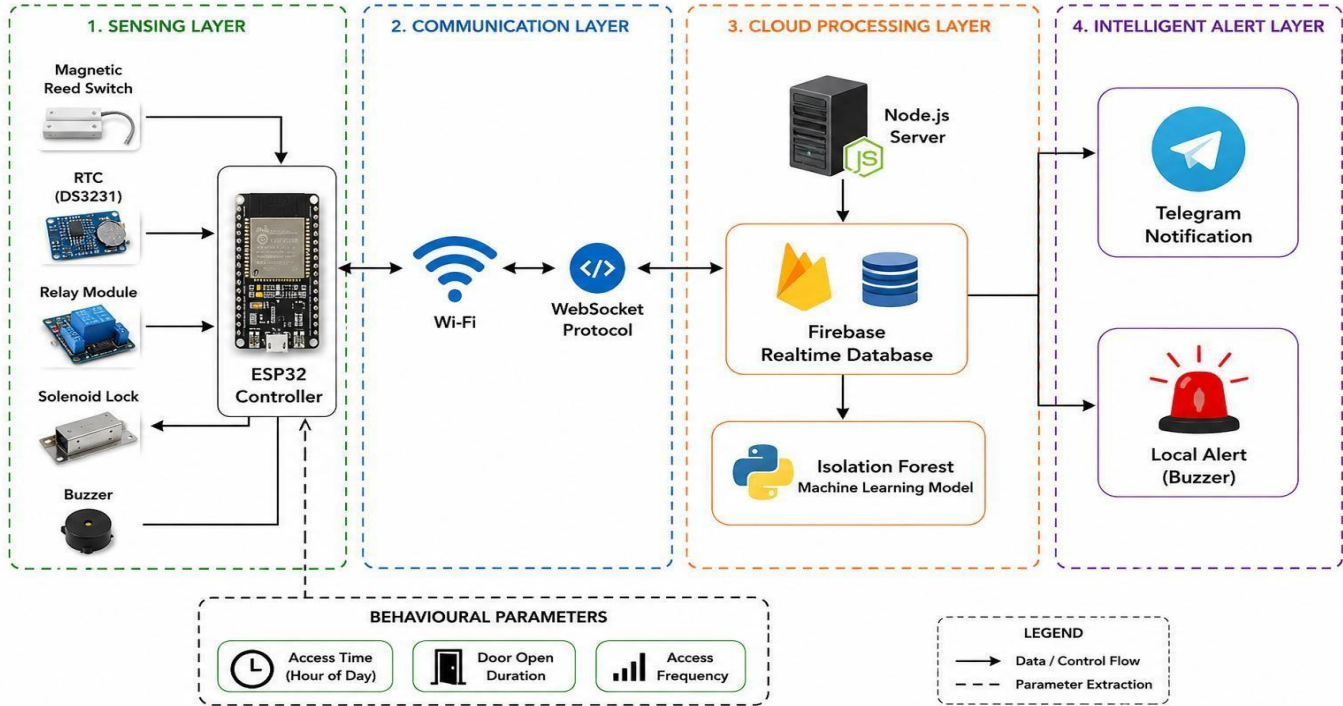


Figure 1. Overall System Architecture of the Proposed Smart Storage Behavioural Intelligence System

5. Methodology

The proposed Smart Storage Behavioural Intelligence System follows a structured methodology that combines embedded hardware, cloud computing, and machine learning to monitor storage access and identify abnormal user behaviour. The overall workflow consists of data acquisition, cloud communication, behavioural analysis, anomaly detection, and real-time alert generation.

A. Hardware Components

The hardware module is built around the ESP32 microcontroller, which serves as the central processing unit of the system. A magnetic reed switch is used to detect the opening and closing of the storage door. The DS3231 Real-Time Clock (RTC) records the exact time of every access event, enabling time-based behavioural analysis. A relay module controls the solenoid lock for secure access, while a buzzer provides an immediate local alert whenever suspicious activity is detected.

B. Software Components

The software architecture consists of Arduino IDE for ESP32 firmware development, a Node.js server for handling WebSocket communication, Firebase Realtime Database for cloud storage, Python for behavioural analysis, and the Telegram Bot API for real-time user notifications. The Node.js server acts as a bridge between the ESP32 and the cloud database while also interacting with the machine learning model for anomaly prediction.

C. Working Principle

Whenever the storage door is accessed, the ESP32 records the access hour using the RTC, calculates the door open duration, and updates the access frequency. These behavioural parameters are transmitted to the Node.js server through a WebSocket connection and stored in Firebase Realtime Database.

6. Proposed System

Algorithm 1: Smart Storage Behavioural Intelligence System

Input: Door status, Access time, Door open duration, Access frequency

Output: Normal Access / Anomaly Detected with Real-Time Alert

1. Initialize the ESP32 microcontroller.
2. Establish Wi-Fi connection.
3. Connect to the Node.js server using the WebSocket protocol.
4. Initialize the DS3231 Real-Time Clock (RTC).
5. Configure the magnetic reed switch, relay module, solenoid lock, and buzzer.
6. Monitor the storage door continuously.
7. Detect a door opening event using the reed switch.
8. Record the access hour from the RTC.
9. Measure the door open duration.
10. Update the access frequency count.
11. Create a behavioural feature vector consisting of {Access Hour, Duration, Frequency}.
12. Transmit the behavioural data to the Node.js server via WebSocket.
13. Store the received data in Firebase Realtime Database.
14. Retrieve historical behavioural records from Firebase.
15. Train or update the Isolation Forest model using historical records.
16. Evaluate the latest access event using the trained model.
17. If the prediction is Normal, store the event and continue monitoring.
18. If the prediction is Anomaly, send a Telegram notification to the authorized user.
19. Transmit an alert message to the ESP32.
20. Activate the buzzer and continue monitoring for subsequent access events.

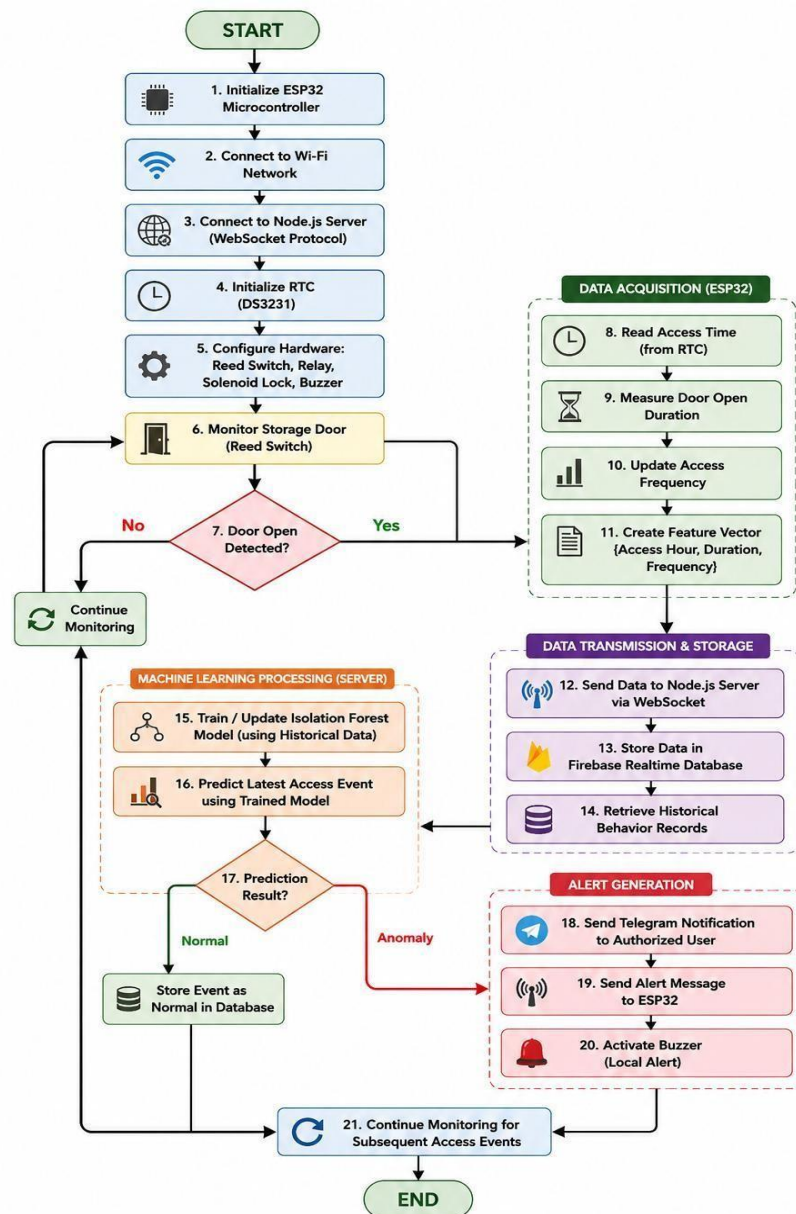


Figure 2. Flowchart of the Proposed System

7. Mathematical Model

The proposed Smart Storage Behavioural Intelligence System utilizes behavioural parameters such as access hour, door open duration, and access frequency to distinguish normal and abnormal storage access events. These behavioural features are extracted from each access event and analysed using the Isolation Forest machine learning algorithm.

A. Behavioural Feature Vector

Each storage access event is represented by a behavioural feature vector:

$$\mathbf{X} = \{\mathbf{H}, \mathbf{D}, \mathbf{F}\}$$

where:

- H = Access Hour
- D = Door Open Duration (seconds)
- F = Access Frequency

B. Behavioural Baseline

The normal behavioural profile is obtained from historical storage access records stored in the Firebase Realtime Database. The average values of each behavioural parameter are calculated as follows:

$$\mathbf{H}_{avg} = (\sum \mathbf{H}_i) / N$$

$$\mathbf{D}_{avg} = (\sum \mathbf{D}_i) / N$$

$$\mathbf{F}_{avg} = (\sum \mathbf{F}_i) / N$$

where:

- N = Total number of historical records
- $\mathbf{H}_i$, $\mathbf{D}_i$, $\mathbf{F}_i$ represent the behavioural parameters of the i th access event.

C. Behavioural Deviation

The deviation between the current access event and the learned behavioural profile is determined using

$$\Delta \mathbf{H} = |\mathbf{H} - \mathbf{H}_{avg}|$$

$$\Delta \mathbf{D} = |\mathbf{D} - \mathbf{D}_{avg}|$$

$$\Delta \mathbf{F} = |\mathbf{F} - \mathbf{F}_{avg}|$$

Higher deviation values indicate greater variation from normal user behaviour.

D. Isolation Forest Prediction

The behavioural feature vector is supplied to the Isolation Forest model for anomaly detection.

$$S(X) = \text{IsolationForest}(X)$$

where $S(X)$ represents the prediction score generated by the trained machine learning model.

- If $S(X) > 0$, the access event is classified as Normal.
- If $S(X) < 0$, the access event is classified as an Anomaly.

E. Alert Generation

The final decision rule is expressed as:

If $S(X) > 0$

Storage Access = Normal

Else

Storage Access = Anomaly

Whenever an anomaly is detected, the Node.js server immediately sends a Telegram notification to the authorized user and simultaneously transmits an alert message to the ESP32 through the WebSocket connection. The ESP32 then activates the buzzer to provide an immediate local security alert while the access record is stored in the Firebase Realtime Database for future behavioural analysis.

8. Implementation

A. Hardware Implementation

The hardware implementation of the proposed Smart Storage Behavioural Intelligence System is centered around the ESP32 microcontroller, which functions as the primary processing unit. The ESP32 interfaces with a magnetic reed switch for door status detection, a DS3231 Real-Time Clock (RTC) for recording accurate access timestamps, a relay-controlled solenoid lock for secure access control, and a buzzer for local alarm generation. A push button is incorporated to provide authorized manual access to the storage unit. The relay module operates the solenoid lock using an external 12 V power supply, while the ESP32 manages all sensing and control operations. Figure 3 illustrates the complete hardware circuit and interconnection of all system components.

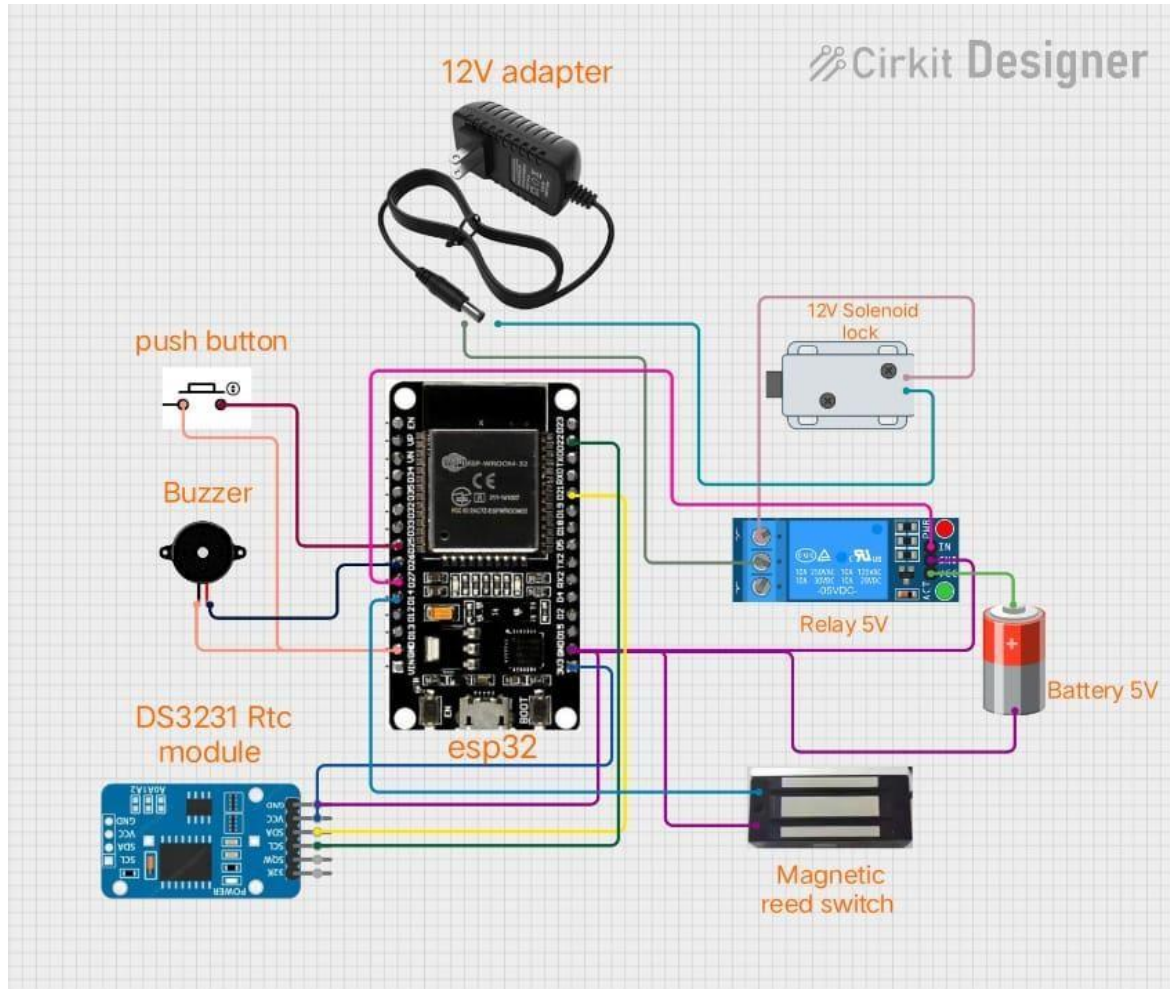


Figure 3. Overall Circuit Diagram of the Proposed Smart Storage Behavioural Intelligence System

B. Software implementation

The software architecture integrates embedded programming, cloud communication, and machine learning to provide intelligent storage monitoring. The ESP32 firmware was developed using Arduino IDE and communicates with a Node.js server through the WebSocket protocol over a Wi-Fi network. Behavioural parameters, including access hour, door open duration, and access frequency, are transmitted to the server and stored in Firebase Realtime Database. A Python-based Isolation Forest model analyses historical and real-time access records to identify abnormal behavioural patterns. Whenever an anomaly is detected, the server generates a Telegram notification for the authorized user and simultaneously sends an alert command to the ESP32, which activates the buzzer.

C. Experimental Setup

The proposed system was experimentally evaluated using multiple storage access events under normal and abnormal operating conditions. Historical access records representing normal user behaviour were collected and stored in Firebase Realtime Database to train the Isolation Forest model. Experimental observations included variations in access time, door open duration, and access frequency. The ESP32 continuously monitored the storage door and transmitted behavioural data to the Node.js server for analysis. Whenever an abnormal access pattern was identified, the system generated an instant Telegram notification and activated the local buzzer, demonstrating reliable real-time anomaly detection and security alert functionality. Figure 5 shows the developed hardware prototype used for experimental validation.

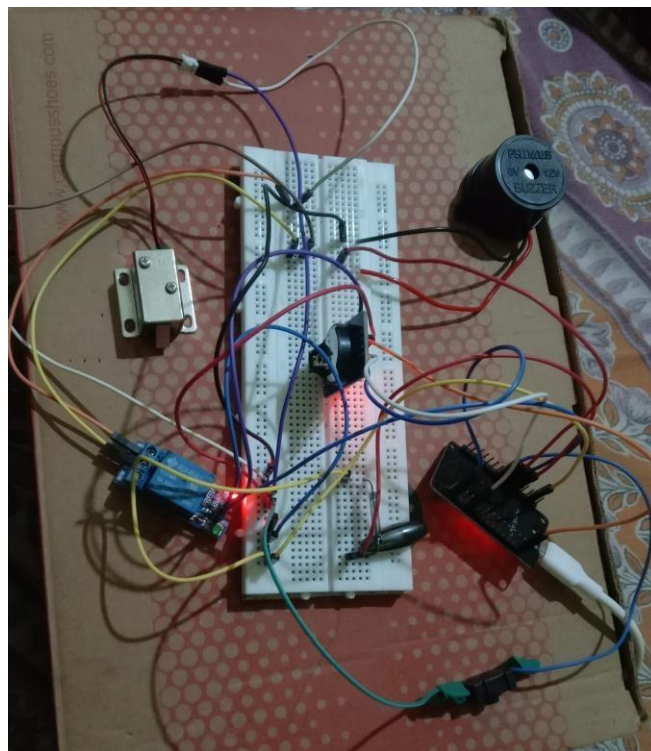


Figure 5. Hardware Prototype of the Proposed Smart Storage System

9. Results And Discussions

The proposed Smart Storage Behavioural Intelligence System was successfully implemented and Experimentally evaluated under various storage access scenarios. The system continuously monitored the Storage door using a magnetic reed switch and recorded behavioural parameters such as access hour, door Open duration, and access frequency. These parameters were transmitted to the Node.js server through the WebSocket protocol and stored in the Firebase Realtime Database for behavioural analysis.

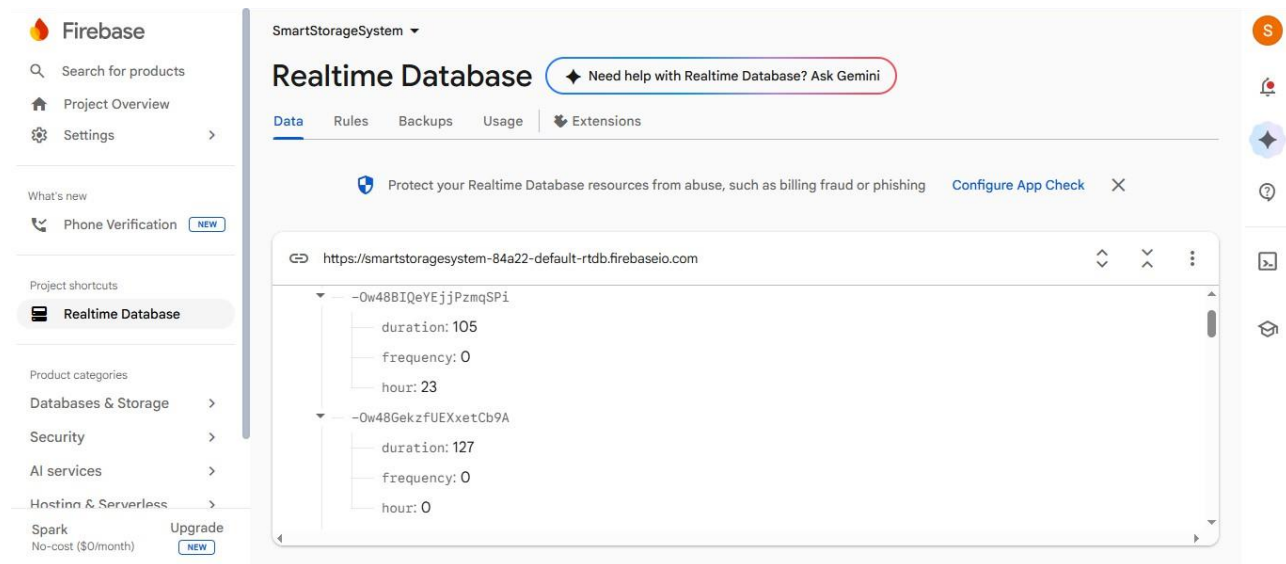


Figure 5. Firebase Realtime Database

Figure 5 illustrates the Firebase Realtime Database used for storing behavioural access records in JSON format. Each access event consists of the access hour, door open duration, and access frequency. The stored historical data serves as the input dataset for training the Isolation Forest machine learning model, enabling the system to learn normal user behaviour and identify abnormal access patterns.

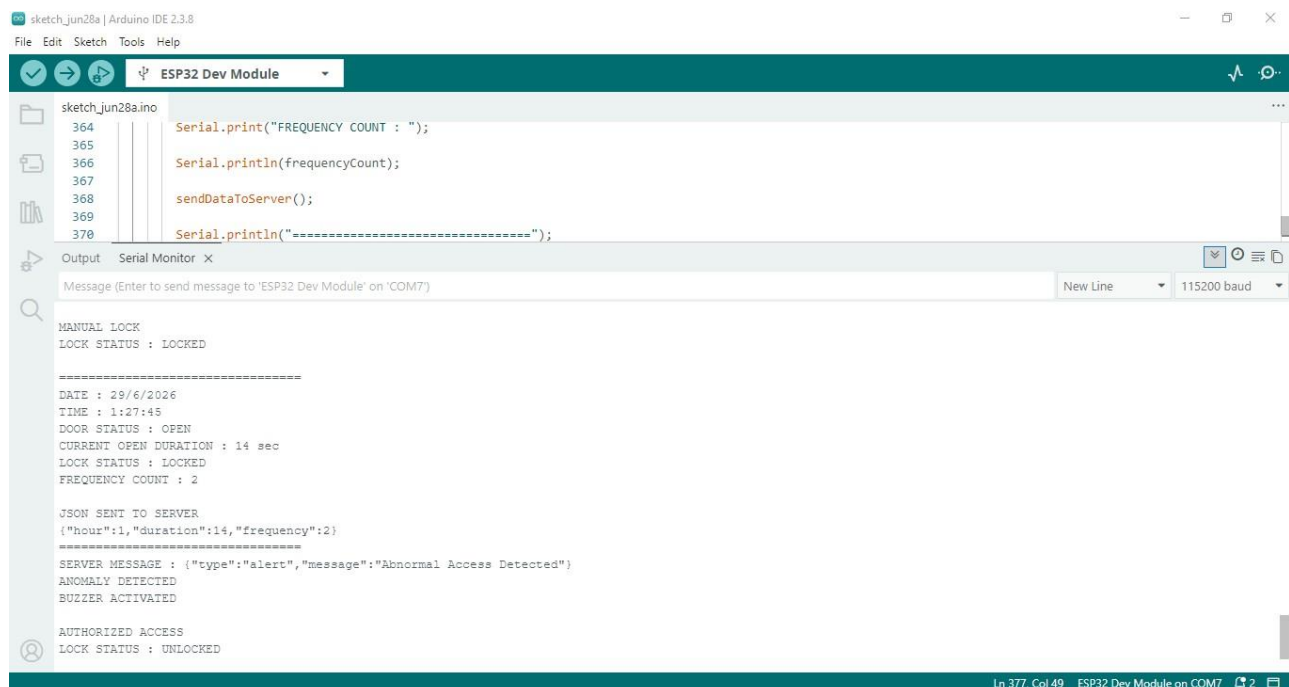
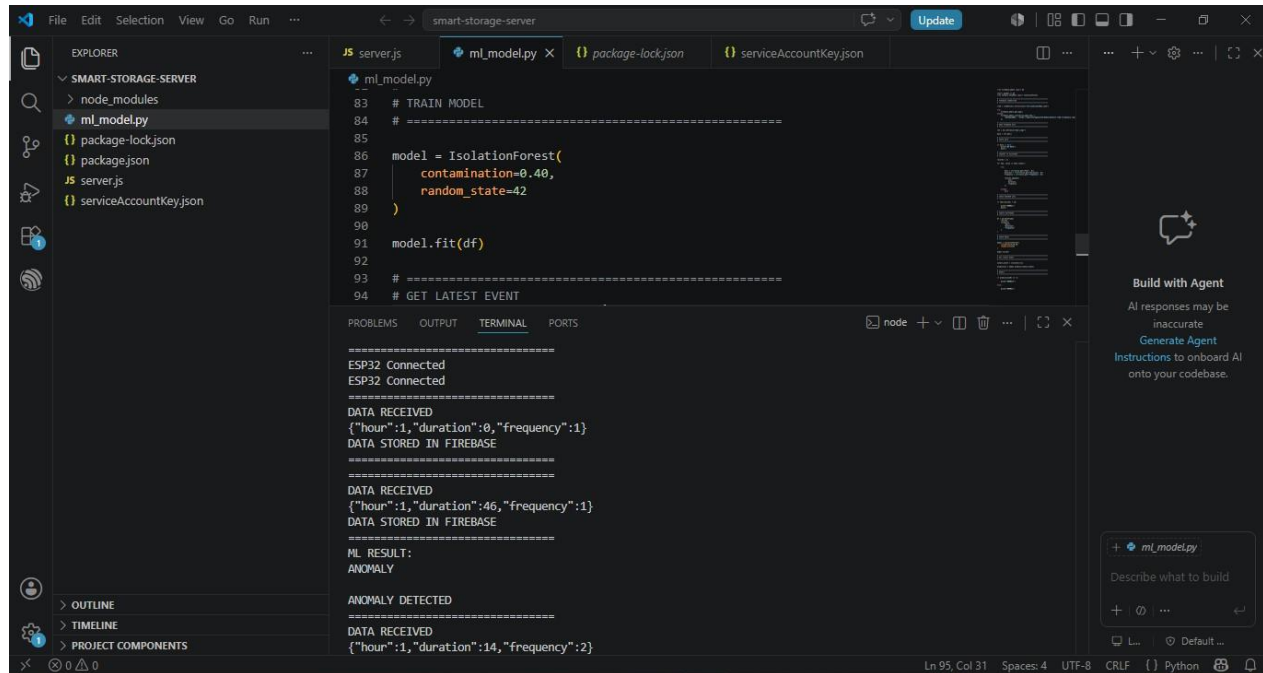


Figure 6. Arduino IDE Serial Monitor

Figure 6 presents the Arduino IDE Serial Monitor displaying the real-time execution of the ESP32 firmware. The serial output confirms successful monitoring of the storage door, acquisition of behavioural parameters, and transmission of JSON-formatted data to the Node.js server through the WebSocket connection. This verifies the proper operation of the embedded hardware and communication module.



The screenshot shows the Node.js server console in VS Code. The terminal output displays the following sequence of events:

```

=====
ESP32 Connected
=====
DATA RECEIVED
{"hour":1,"duration":0,"frequency":1}
DATA STORED IN FIREBASE
=====
DATA RECEIVED
{"hour":1,"duration":46,"frequency":1}
DATA STORED IN FIREBASE
=====
ML RESULT:
ANOMALY
=====
ANOMALY DETECTED
=====
DATA RECEIVED
{"hour":1,"duration":14,"frequency":2}
=====

```

The code in the background shows the training of an Isolation Forest model with parameters: `contamination=0.40` and `random_state=42`.

Figure 7. Node.js Server Output

Figure 7 shows the Node.js server console after receiving behavioural data from the ESP32. The server successfully establishes WebSocket communication, stores the received records in Firebase Realtime Database, and executes the Isolation Forest machine learning model for anomaly prediction. The server continuously processes incoming access events and determines whether the observed behaviour is normal or anomalous.

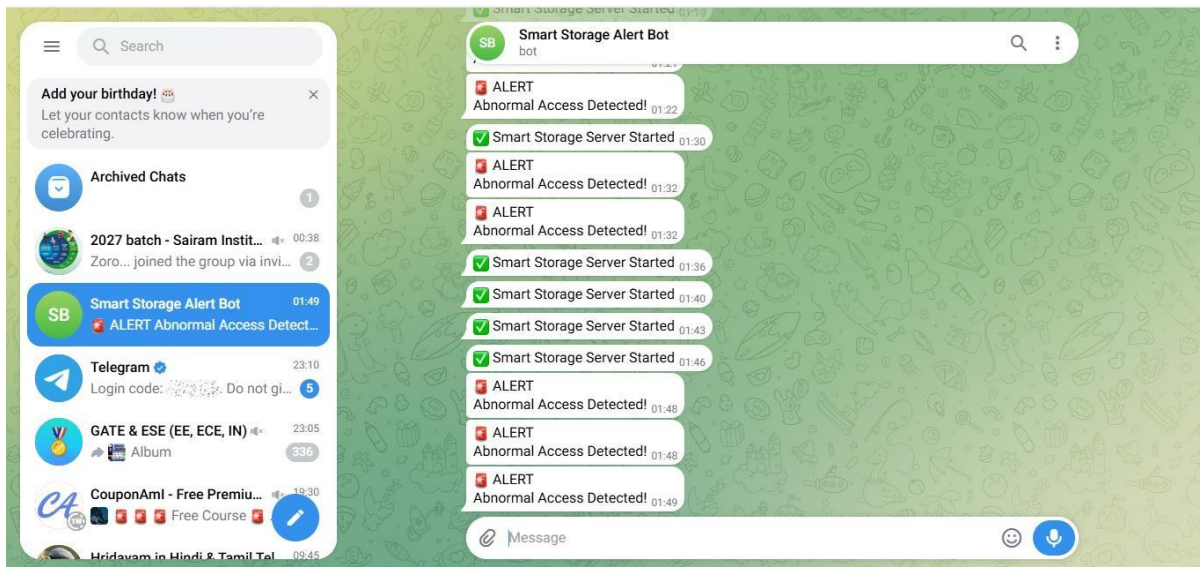


Figure 8. Telegram Alert Notification

Figure 8 illustrates the Telegram notification generated whenever an abnormal storage access event is detected. Upon receiving an anomalous prediction from the Isolation Forest model, the Node.js server immediately sends a notification to the authorized user and simultaneously transmits an alert message to the ESP32, which activates the buzzer for local warning. The experimental results demonstrate that the proposed Smart Storage Behavioural Intelligence System successfully integrates embedded hardware, cloud computing, and machine learning to provide reliable real-time behavioural monitoring and intelligent storage security.

10. Conclusion

This paper presented a Smart Storage Behavioural Intelligence System Using IoT and Machine Learning for Anomaly Detection and Real-Time Security Alerts. The proposed system successfully integrates an ESP32 microcontroller, magnetic reed switch, relay-controlled solenoid lock, DS3231 Real-Time Clock (RTC), Firebase Realtime Database, WebSocket communication, and the Isolation Forest machine learning algorithm to provide intelligent storage security. Behavioural parameters including access hour, door open duration, and access frequency were continuously monitored and analysed to distinguish normal and abnormal storage access events.

The experimental implementation demonstrated reliable real-time communication between the embedded hardware, cloud database, and machine learning model. Historical access records stored in Firebase enabled the Isolation Forest algorithm to learn normal user behaviour and accurately identify anomalous access patterns. Whenever abnormal activity was detected, the system successfully generated instant Telegram notifications and activated a local buzzer, thereby providing both remote and on-site security alerts.

The proposed Smart Storage Behavioural Intelligence System offers a cost-effective, scalable, and intelligent security solution for residential lockers, office cabinets, industrial storage units, and other smart storage applications. By combining IoT technology with behavioural intelligence, the system enhances storage security beyond conventional authentication methods and provides an adaptive approach for real-time anomaly detection and user notification.

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